

HOME PAGE ▾**TOPICS ▾**

Controlling Motors
 Analogue Electronics
 Digital Electronics
 Components Explained
 Practical Matters
 Model Studies

FEATURES ▾**CIRCUITS SHOP ▾****Search The Site...**

Look for

In Articles

Go >>

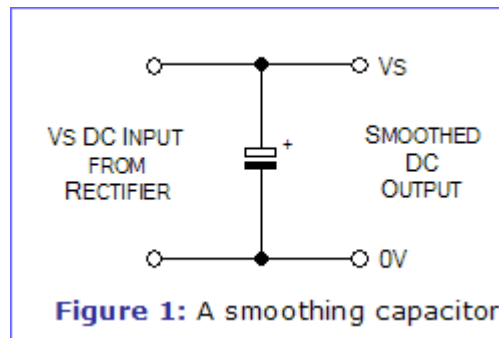
Things To Do...

Print this article
 Add this article to
 your Favourites
 Read this topic's
 Message Board

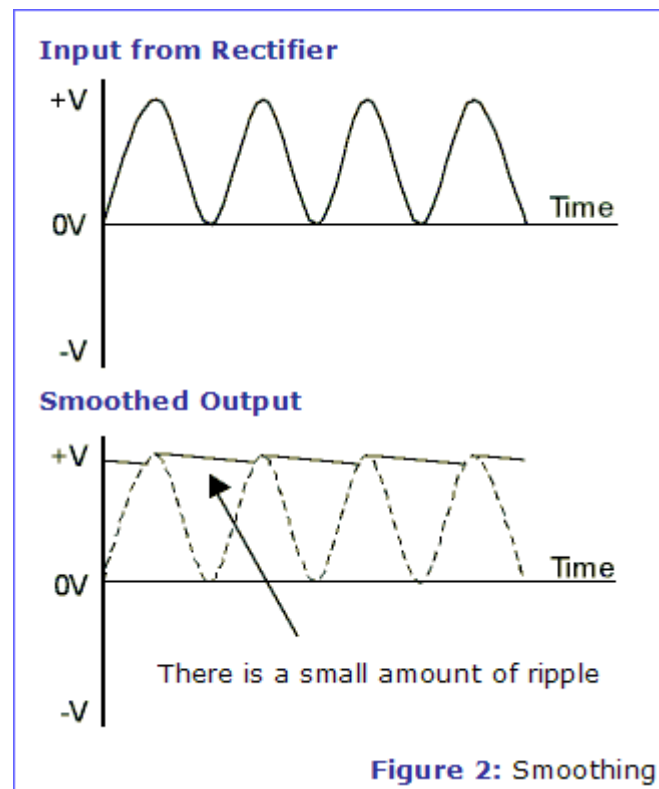
Smoothing

By Tim Surtell

Most circuits will require 'smoothing' of the DC output of a rectifier, and this is a simple matter since it involves only one capacitor, as shown in figure 1.

**Further Information** ► [Capacitors](#)

The output waveform in figure 2 shows how smoothing works. During the first half of the voltage peaks from the rectifier, when the voltage increases, the capacitor charges up. Then, while the voltage decreases to zero in the second half of the peaks, the capacitor releases its stored energy to keep the output voltage as constant as possible. Such a capacitor is called a 'smoothing' or 'reservoir' capacitor when it is used in this application.



ELECTRONICS IN MECCANO

Ads by Goooooogle

Electronic Circuit Boards

Find Electronic Circuit
 Board Jobs Submit Circuit
 Board RFQs
mfgquote.com

Telecom Power Systems

48V & 24V Systems to
 10,000+ A for CO, cell
 site, & cabinet power
www.power.tycoelectronics.com

Max Perf PC Power Supply

Turbo-Cool 1000W. Buy
 The Ultimate Computer
 Power Supply
www.PCPower.com

Vicor power supply designs

DC-DC and AC-DC power
 modules for all your
 application designs
www.vicoreurope.com

Electronics Projects

Browse a Good collection
 of Free Electronics
 Projects
www.reloadedlabs.com

[Advertise on this site](#)

Ripple

If the voltage peaks from the rectifier were not continually charging up the capacitor, it would eventually discharge and the output voltage would decrease all the way down to 0V. The discharging that does occur between peaks gives rise to a small 'ripple' voltage. The amount of ripple is affected by a combination of three factors:

- **The value of the capacitor.** The larger the capacitor value, the more charge it can store, and the slower it will discharge. Therefore, smoothing capacitors are normally electrolytic capacitors with values over 470µF.
- **The amount of current used by the circuit.** If the circuit connected to the power supply takes a lot of current, the capacitor will discharge more quickly and there will be a higher ripple voltage.
- **The frequency of the peaks.** The more frequent the voltage peaks from the rectifier, the more often the capacitor will be charged, and the lower the ripple voltage will be.

If you want to calculate the ripple voltage, you can use this formula...

$$V_r = \frac{I}{CF}$$

...where V_r is the ripple voltage in Volts, I is the current taken by the circuit in Amps, C is the value of the smoothing capacitor in Farads, and F is the frequency of the peaks from the full-wave rectifier, in Hertz. This frequency will be double the normal mains frequency, i.e. 100Hz in the case of the UK mains supply, or 120Hz in the case of the US mains supply.

The ripple voltage should not be more than 10% of V_s - if it is, increase the value of the smoothing capacitor.

Lots of circuits will work fine from a smoothed power supply, but some must have a completely regular supply with no ripple voltage. The next article in this series discusses regulators which can provide this regular power supply.



- Related Articles**
- ▶ [Building a Power Supply](#) - Part 1 of this series
 - ▶ [The Transformer](#) - Part 2 of this series
 - ▶ [The Rectifier](#) - Part 3 of this series
 - ▶ [The Regulator](#) - Part 5 of this series



Shopping List

Part Description	Quantity Required	Maplin Code	Unit Price	Total Price
470µF 63V Radial Electrolytic Capacitor	1	VH48C	£0.72	£0.72
Grand Total				£0.72

► [Full printer-friendly parts list](#)

Order the above items from Maplin via our Circuits Shop and help support the *Electronics in Meccano* website, without affecting the price you pay! Just click on an item for information about it and/or to add it to your Maplin order.

Orders can also be placed by telephoning Maplin on 0870 429 6000, or visiting their website at www.maplin.co.uk. Prices quoted are taken from the Autumn/Winter 2006/7 Maplin catalogue and include VAT at 17.5%.

Confused???

Post your questions and comments about this article to the [Analogue Electronics Message Board](#).

We also have many other [message boards](#) for your comments, advice and suggestions.

Free Email Updates!

Join [yourEiM](#) and receive email when your favourite articles and topics are updated. [Click here](#) to sign up now!

Print This Article

Article Information

Topic: [Analogue Electronics](#) | Created: 02/05/2003 | Last modified: 09/02/2007 | First published in EiM: Issue 4 (June 1999)

[Top of Page](#) | [Homepage](#) | [About](#) | [Search](#) | [Topics](#) | [Features](#) | © 1998 - 2007 Tim Surtell
[Circuits Shop](#) | [yourEiM](#)